KAIZEN MATHS

Lesson starter

Advanced Differentiation

Implicit Differentiation

Name _____

S1

Given $x^2 + 2xy + 4y^2 = 14$, find $\frac{dy}{dx}$.

S2

For $x^2y + y^2 = 52$, find $\frac{dy}{dx}$ at $(3, 4)$.

S3

For $3x^2 + 2y^2 = 21$, find $\frac{d^2y}{dx^2}$.

S4

For $x^2 + 4xy + 5y^2 = 90$, find the tangent line at $(3, 3)$.

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Exit ticket

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E1

Given $4x^2 + 3y^2 = 63$ and $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$ at $x = 3$ and $y = 3$.

E2

Given $x^2 + 2xy + 3y^2 = 77$, find $\frac{dy}{dx}$.

E3

For $xy = 15$, find $\frac{dy}{dx}$ at $(3, 5)$.

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Teacher answers & working

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S1

Given $x^2 + 2xy + 4y^2 = 14$, find $\frac{dy}{dx}$.

Answer:

$$\frac{dy}{dx} = -\frac{2x + 2y}{2x + 8y}$$

Working:

Differentiate each term. Remember: y depends on x.

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx}(2xy) = 2\left(x\frac{dy}{dx} + y\right)$$

$$\frac{d}{dx}(4y^2) = 8y\frac{dy}{dx}$$

$$\frac{d}{dx}(14) = 0$$

Put the derivatives together.

$$2x + 2\left(x\frac{dy}{dx} + y\right) + 8y\frac{dy}{dx} = 0$$

Expand the brackets.

$$2x + 2x\frac{dy}{dx} + 2y + 8y\frac{dy}{dx} = 0$$

Move the terms without dy/dx.

$$2x\frac{dy}{dx} + 8y\frac{dy}{dx} = -2x - 2y$$

Factor out dy/dx.

$$(2x + 8y)\frac{dy}{dx} = -(2x + 2y)$$

Divide by the coefficient of dy/dx.

$$\frac{dy}{dx} = -\frac{2x + 2y}{2x + 8y}$$

Teacher answers & working

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S2

For $x^2y + y^2 = 52$, find $\frac{dy}{dx}$ at $(3, 4)$.

Answer:

$$\frac{dy}{dx} = \frac{-24}{17}$$

Working:

Differentiate each term. Remember: y depends on x.

$$\frac{d}{dx}(x^2y) = x^2\frac{dy}{dx} + y(2x)$$

$$\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$$

$$\frac{d}{dx}(52) = 0$$

$$2xy + x^2\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

Move the terms without dy/dx.

$$x^2\frac{dy}{dx} + 2y\frac{dy}{dx} = -2xy$$

Factor out dy/dx.

$$(x^2 + 2y)\frac{dy}{dx} = -2xy$$

Divide by the coefficient of dy/dx.

$$\frac{dy}{dx} = -\frac{2xy}{x^2 + 2y}$$

Substitute x = 3, y = 4.

$$m = -\frac{2(3)(4)}{(3)^2 + 2(4)}$$

$$m = -\frac{24}{17}$$

$$m = \frac{-24}{17}$$

Teacher answers & working

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S3

For $3x^2 + 2y^2 = 21$, find $\frac{d^2y}{dx^2}$.

Answer:

$$\frac{d^2y}{dx^2} = -\frac{63}{4y^3}$$

Working:

Differentiate each term. Remember: y depends on x.

$$\frac{d}{dx}(3x^2) = 6x$$

$$\frac{d}{dx}(2y^2) = 4y\frac{dy}{dx}$$

$$\frac{d}{dx}(21) = 0$$

$$6x + 4y\frac{dy}{dx} = 0$$

Move the terms without dy/dx.

$$4y\frac{dy}{dx} = -6x$$

Divide by the coefficient of dy/dx.

$$\frac{dy}{dx} = -\frac{6x}{4y}$$

Cancel the common factor 2.

$$\frac{dy}{dx} = -\frac{3x}{2y}$$

Differentiate using the quotient rule.

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$u = -3x, \quad u' = -3$$

$$v = 2y, \quad v' = 2\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{(-3)(2y) - (-3x)(2dy/dx)}{4y^2}$$

Substitute the first derivative.

$$\frac{d^2y}{dx^2} = \frac{-6y + 6x(-3x/(2y))}{4y^2}$$

$$\frac{d^2y}{dx^2} = \frac{-6y - 9x^2/y}{4y^2}$$

Multiply numerator and denominator by y.

$$\frac{d^2y}{dx^2} = -\frac{6y^2 + 9x^2}{4y^3}$$

Factor the numerator.

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S3 (continued)

$$\frac{d^2y}{dx^2} = -\frac{3(3x^2 + 2y^2)}{4y^3}$$

Use $3x^2 + 2y^2 = 21$.

$$\frac{d^2y}{dx^2} = -\frac{3(21)}{4y^3}$$

$$\frac{d^2y}{dx^2} = -\frac{63}{4y^3}$$

Teacher answers & working

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S4

For $x^2 + 4xy + 5y^2 = 90$, find the tangent line at $(3, 3)$.

Answer:

$3x + 7y - 30 = 0$

Working:

Differentiate each term. Remember: y depends on x.

$\frac{d}{dx}(x^2) = 2x$

$\frac{d}{dx}(4xy) = 4\left(x\frac{dy}{dx} + y\right)$

$\frac{d}{dx}(5y^2) = 10y\frac{dy}{dx}$

$\frac{d}{dx}(90) = 0$

Put the derivatives together.

$2x + 4\left(x\frac{dy}{dx} + y\right) + 10y\frac{dy}{dx} = 0$

Expand the brackets.

$2x + 4x\frac{dy}{dx} + 4y + 10y\frac{dy}{dx} = 0$

Move the terms without dy/dx.

$4x\frac{dy}{dx} + 10y\frac{dy}{dx} = -2x - 4y$

Factor out dy/dx.

$(4x + 10y)\frac{dy}{dx} = -(2x + 4y)$

Divide by the coefficient of dy/dx.

$\frac{dy}{dx} = -\frac{2x + 4y}{4x + 10y}$

Substitute x = 3, y = 3.

$m = -\frac{2(3) + 4(3)}{4(3) + 10(3)}$

$m = -\frac{18}{42}$

$m = \frac{-3}{7}$

Use point-gradient form.

$y - y_1 = m(x - x_1)$

$y - 3 = \frac{-3}{7}(x - 3)$

Clear the denominator.

$7(y - 3) = -3(x - 3)$

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S4 (continued)

Expand.

$7y - 21 = -3x - (-9)$

Collect all terms on one side.

$3x + 7y - 30 = 0$

E1

Given $4x^2 + 3y^2 = 63$ and $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$ at $x = 3$ and $y = 3$.

Answer:

$\frac{dy}{dt} = -4$

Working:

Differentiate each term with respect to time.

$\frac{d}{dt}(4x^2) = 8x\frac{dx}{dt}$

$\frac{d}{dt}(3y^2) = 6y\frac{dy}{dt}$

$\frac{d}{dt}(63) = 0$

$8x\frac{dx}{dt} + 6y\frac{dy}{dt} = 0$

Move the dx/dt term.

$6y\frac{dy}{dt} = -8x\frac{dx}{dt}$

Divide by the coefficient of dy/dt.

$\frac{dy}{dt} = -\frac{8x}{6y}\frac{dx}{dt}$

Cancel the common factor 2.

$\frac{dy}{dt} = -\frac{4x}{3y}\frac{dx}{dt}$

Substitute the given values.

$\frac{dy}{dt} = -\frac{4(3)}{3(3)}(3)$

$\frac{dy}{dt} = -\frac{36}{9}$

$\frac{dy}{dt} = -4$

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E2

Given $x^2 + 2xy + 3y^2 = 77$, find $\frac{dy}{dx}$.

Answer:

$$\frac{dy}{dx} = -\frac{2x + 2y}{2x + 6y}$$

Working:

Differentiate each term. Remember: y depends on x.

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx}(2xy) = 2\left(x\frac{dy}{dx} + y\right)$$

$$\frac{d}{dx}(3y^2) = 6y\frac{dy}{dx}$$

$$\frac{d}{dx}(77) = 0$$

Put the derivatives together.

$$2x + 2\left(x\frac{dy}{dx} + y\right) + 6y\frac{dy}{dx} = 0$$

Expand the brackets.

$$2x + 2x\frac{dy}{dx} + 2y + 6y\frac{dy}{dx} = 0$$

Move the terms without dy/dx.

$$2x\frac{dy}{dx} + 6y\frac{dy}{dx} = -2x - 2y$$

Factor out dy/dx.

$$(2x + 6y)\frac{dy}{dx} = -(2x + 2y)$$

Divide by the coefficient of dy/dx.

$$\frac{dy}{dx} = -\frac{2x + 2y}{2x + 6y}$$

Teacher answers & working

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E3

For $xy = 15$, find $\frac{dy}{dx}$ at $(3, 5)$.

Answer:

$$\frac{dy}{dx} = \frac{-5}{3}$$

Working:

Differentiate each term. Remember: y depends on x.

$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y(1)$$

$$\frac{d}{dx}(15) = 0$$

$$x\frac{dy}{dx} + y = 0$$

Move the terms without dy/dx.

$$x\frac{dy}{dx} = -y$$

Divide by the coefficient of dy/dx.

$$\frac{dy}{dx} = -\frac{y}{x}$$

Substitute x = 3, y = 5.

$$m = -\frac{(5)}{(3)}$$

$$m = -\frac{5}{3}$$

$$m = \frac{-5}{3}$$